

PD Group:

Name:

ANDERSON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS

BIOLOGY
Higher 2

9747/03

Paper 3 Applications Paper

8 SEPTEMBER 2008 Monday

1 hour 30 minutes**READ THESE INSTRUCTIONS FIRST**

Write your name and PD group on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graph or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

All working for numerical answers must be shown.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use**1****2****3****4****Total**This document consists of **9** printed pages.**[Turn over]**

Section A

Answer **all** the questions in this section.

- 1 Analysis of restriction fragment length polymorphism (RFLP) fragments is an important tool in molecular biology.

(a) (i) What are RFLP fragments?

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[1]

(ii) Why do two individuals express different patterns of RFLP fragments?

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[2]

(iii) State two uses of RFLP fragments to society.

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[2]

- (b) RFLP analysis is used within Southern blotting to check for the presence of a disease-causing mutant allele in a DNA sample.

Fig 1.1 shows the results of Southern blotting for the gene of interest.

Inheritance of RFLP markers

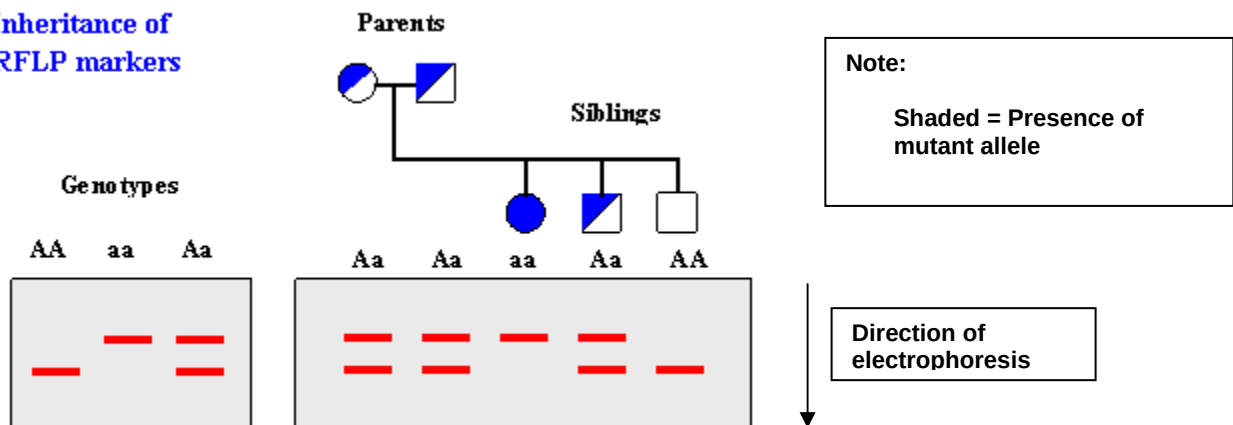


Fig.1.1

- (i) Outline the process of Southern blotting.

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.....[5]

- (ii) What is the mode of inheritance of this disease?

.....[1]

- (iii) Explain for the bands found in the individuals.

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.....[4]

[Total: 15]

- 2 (a) Cystic fibrosis is a genetic disorder caused by a recessive mutation. It is characterized by malfunction of pancreas and abnormal secretion from the lungs and is fatal if symptoms are untreated.

However, heterozygotes for the most common mutant allele causing cystic fibrosis (CF) may be more resistant to bacterial infections of the gut than homozygotes for the normal allele.

- (i) Explain how such resistance could affect the frequency of the mutant allele in a population.

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.....[2]

The bacterium *Salmonella typhi*, which causes typhoid fever in humans, will infect mouse gut cells. Mouse gut epithelium cells were engineered to express either the normal allele or the mutant allele of the gene responsible for cystic fibrosis. Three strains of *S. typhi* were incubated with cells expressing the mutant CF allele and cells expressing the normal allele. Fig. 2 shows the numbers of bacteria taken up by these cells.

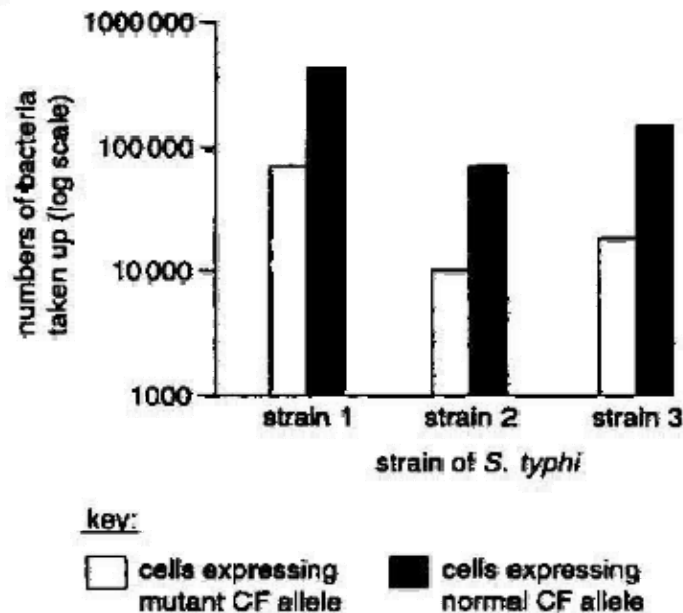


Fig.2.1

- (ii) With reference to Fig.2.1, compare the effects of the normal and mutant alleles for cystic fibrosis on the number of bacteria taken up by the mouse gut cells.

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.....[3]

- (iii) Suggest why the expression of the cystic fibrosis gene affects the number of bacteria taken up.

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- (b) Thalassaemia is an inherited autosomal recessive blood disease. In β -thalassaemia, the genetic defect results in reduced rate of synthesis of the β globin chains that make up hemoglobin.

Many cases of the genetic disease β -thalassaemia result from a large deletion in the gene for β globin.

- (i) Explain how electrophoresis allows an allele with a large deletion to be distinguished from the normal allele.

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.....[2]

A potential gene therapy for β -thalassaemia involves adding the normal, dominant allele for β globin to the patient's bone marrow cells.

- (ii) Explain why it is theoretically easier to perform gene therapy when a mutant allele is recessive than when it is dominant.

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.....[3]

Such gene therapy has been tested in mice. Bone marrow cells carrying the normal allele for human β globin were implanted both in normal mice and in mice heterozygous for the deletion causing β -thalassaemia. Twenty-four weeks later, the percentage of haemoglobin containing two human β globin chains was measured in both types of mouse. The results are shown in Table 2.1.

Type of mouse	Percentage of mouse haemoglobin with two human β globin chains
Normal	13
Heterozygous for the deletion causing β -thalassaemia	24

Table 2.1

- (iii) With reference to Table 2.1, suggest why the human gene is expressed less in normal mice.

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.....[2]

[Total: 15]

3 (a) Transgenic plants are created from genetic engineering to introduce desirable traits for economically important plants.

(i) Define transgenic plants and state an example of one economically important transgenic plant.

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(ii) Give an example of a desirable trait within the context of an economically important plant.

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.....[1]

(iii) Outline how the gene that codes for this desirable trait could have been isolated.

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(iv) Explain the use of Ti plasmid in genetic engineering of plants

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.....[3]

(b) *Staphylococcus aureus*, a gram positive bacterium, synthesizes a large number of extracellular proteins that have been postulated to play a role in bacterial virulence. Researchers discovered that the thermostability of some of these proteins differ between the wild type and genetically mutated *S. aureus*.

Fig. 3.1 shows the rate of protein production by *S. aureus* at various temperatures.

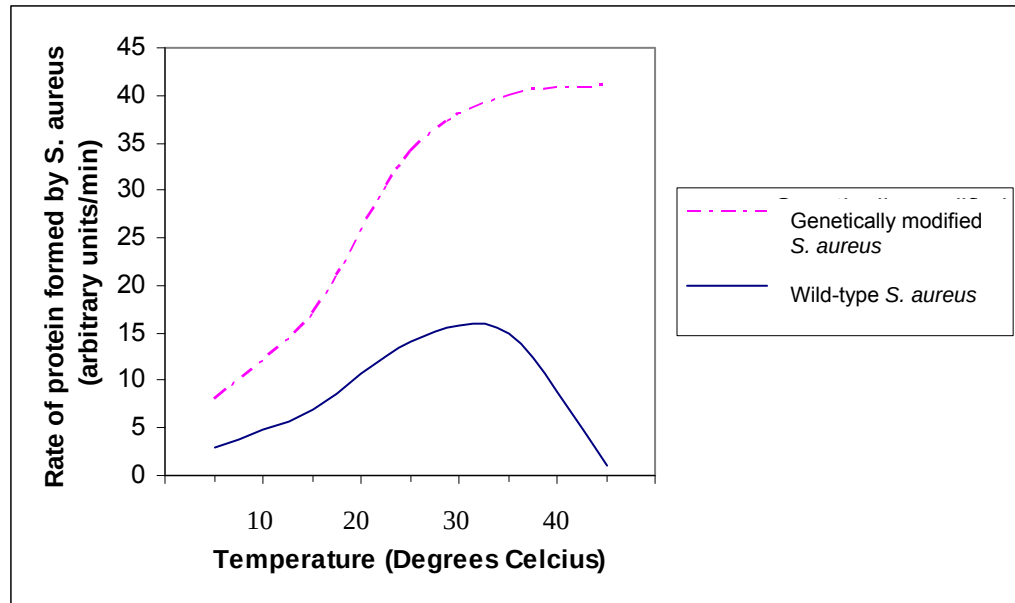


Fig 3.1

- (i) Describe the effects of increasing temperature on the quantity of product formed for the 2 types of bacteria.

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- (ii) Suggest what genetic modifications have occurred to account for the observations in the genetically modified *S. aureus*.

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.....[2]

[Total: 15]

Write your answer to this question on the separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in section **(a)**, **(b)** etc., as indicated in the question.

- 4 **(a)** Describe the characteristics of hematopoietic stem cells. [8]
- (b)** Outline how medically important proteins can be mass produced via genetic engineering. [8]
- (c)** Briefly discuss two societal concerns arising from the Human Genome Project. [4]

[Total: 20]